

Ligjerata 6a

Puna mekanike e lëvizjes translative

$$A = \vec{F} \cdot \vec{s}$$

$$A = F \cdot s \cdot \cos \varphi$$

$$\varphi = 0^\circ, \cos 0^\circ = 1$$

$$A = F \cdot s$$

$$\varphi = 90^\circ, \cos 90^\circ = 0$$

$$A = 0$$

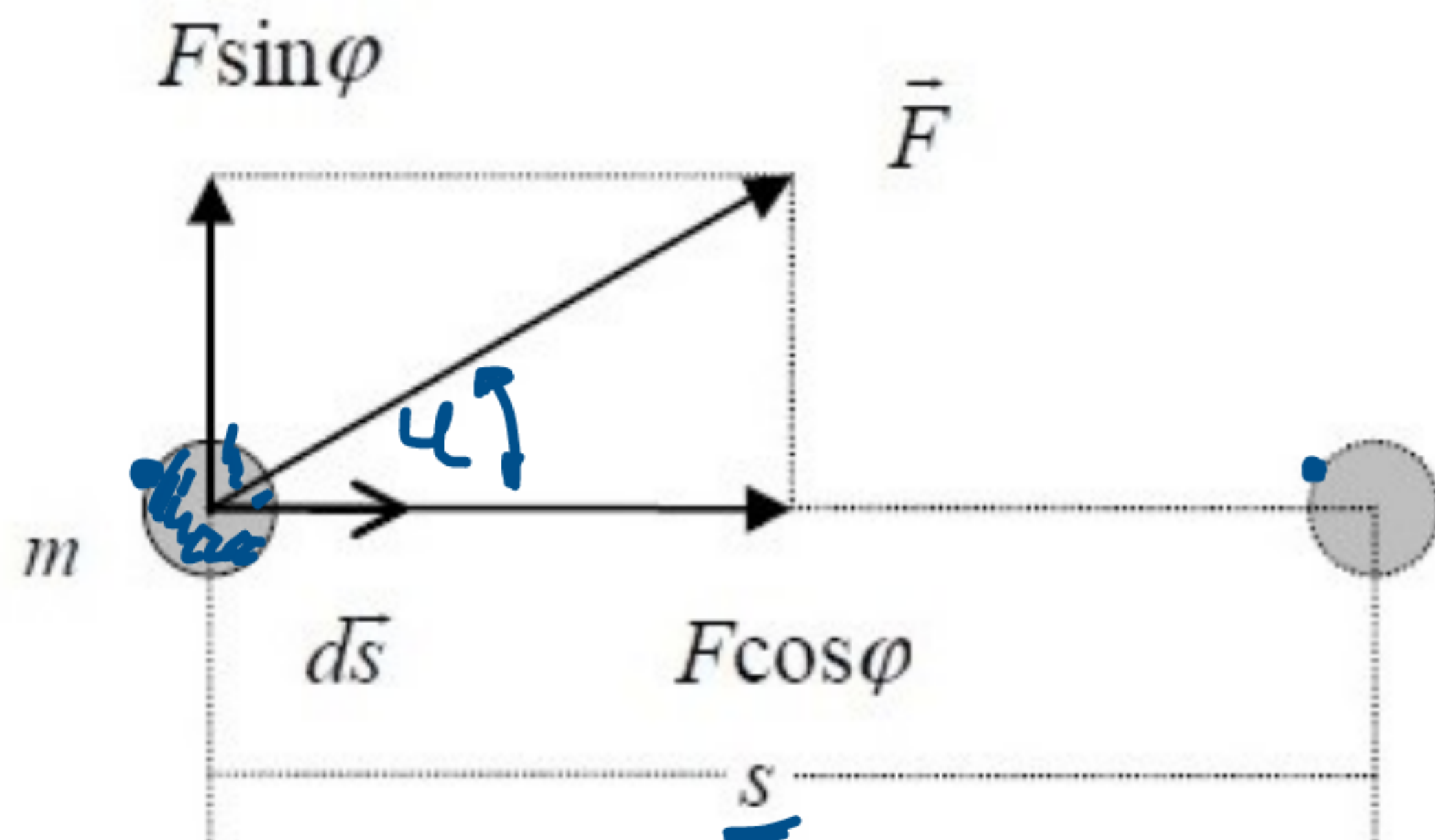
A → puna mekanike

F → forca

s → zhvendosja

ϕ → këndi në mes të drejtimit të forcës dhe zhvendosjes.

$$A = F \cdot s = N \cdot m = J \text{ (Xhuli)}$$



$$\Delta A = F \cdot \Delta s \cdot \cos \varphi$$

$$\Delta t \rightarrow 0$$

$$\lim_{\Delta t \rightarrow 0} \Delta A = F \lim_{\Delta t \rightarrow 0} \Delta s \cdot \cos \varphi$$

$$\int_0^s dA = \int_{s_0}^s F \cdot ds \cdot \cos \varphi, \varphi = 0$$

$$A = F \int_{s_0}^s ds \rightarrow A = \int_{s_0}^s \vec{F} \cdot d\vec{s}$$

